My model is made to predict shot probabilities. I had engineered almost all of my features using the raw tracking data to capture spatial geometry and defensive pressure, most notably known in my code as “defenderratio” to evaluate how closely and tightly a shooter was contested relative to their distance from the hoop. High cardinality features were frequency encoded and low cardinality categories were used for model training “HistGradientBoosteingClassifier. Validated with Stratified 5-fold cross validation to prevent data leakage and Permutation feature importance confirmed that defender_ratio was by far the strongest predictive driver of shot success, outperforming absolute distance and shot type alone.